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Editorial Board

Linear Algebra and its Applications (LAA)

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Dear Editor-in-Chief and Members of the Editorial Board,

On behalf of my co-author, Dr. Sheehan Olver, and myself, I am writing to submit our manuscript titled “*The QR factorization of a banded-plus-semiseparable matrix is computable in linear complexity with Householder reflections*” for consideration for publication in *Linear Algebra and its Applications*.

Banded-plus-semiseparable (BPS) matrices arise naturally in spectral methods for discretizing differential equations, integral equations, and structured matrix inverses. While classical literature, such as foundational works by Vandebril, Van Barel, Eidelman, and others, has extensively studied rank-structured matrices using specially designed sequences of Givens rotations, the direct use of Householder reflections for BPS matrices remains unexplored to our knowledge.

In this work, we present a Householder-based $O(n)$ QR factorization framework that offers distinct practical and theoretical advantages over traditional Givens-rotation approaches. Unlike existing bottom-left Givens rotation schemes, our Householder formulation computes at each stage a structured representation of standard factor matrices natively compatible with LAPACK unblocked QR layouts. Building upon this structured representation, we develop a complete $O(n)$ direct solver incorporating fast applications of $Q^\top$ and backward substitution. The adaptive, partial QR factorizations whose computational cost depends solely on the number of upper-triangularized columns rather than the global matrix size, after an initial $O(n)$ precomputation of a lookup table that can in certain cases be done explicitly. Furthermore, we rigorously establish that the upper triangular factor $T \in \mathbb{R}^{n \times n}$ arising from LAPACK’s Compact WY representation ($Q = I - VTV^\top$) inherently inherits a rigid BPS structure, enabling seamless integration with dense BLAS-3 routines (potentially on GPUs). Finally, for symmetric BPS matrices, we prove that the reverse RQ product strictly preserves the BPS format, establishing structural closure under orthogonal similarity transformations.

Given the manuscript’s strong focus on matrix analysis, structural preservation properties, and fast numerical linear algebra, we believe our theoretical results and algorithms align closely with the scope and readership of LAA.

For the handling editor or potential reviewers, we would like to suggest Alex Townsend, who has extensive expertise in spectral methods and hierarchical linear algebra, areas where BPS matrices arise naturally.

We confirm that this manuscript is original, unpublished, and not under consideration for publication elsewhere. All authors have approved the submission of this manuscript. Thank you very much for your time and consideration.

Sincerely yours,

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